HAZELCAST

SA Technical Brief

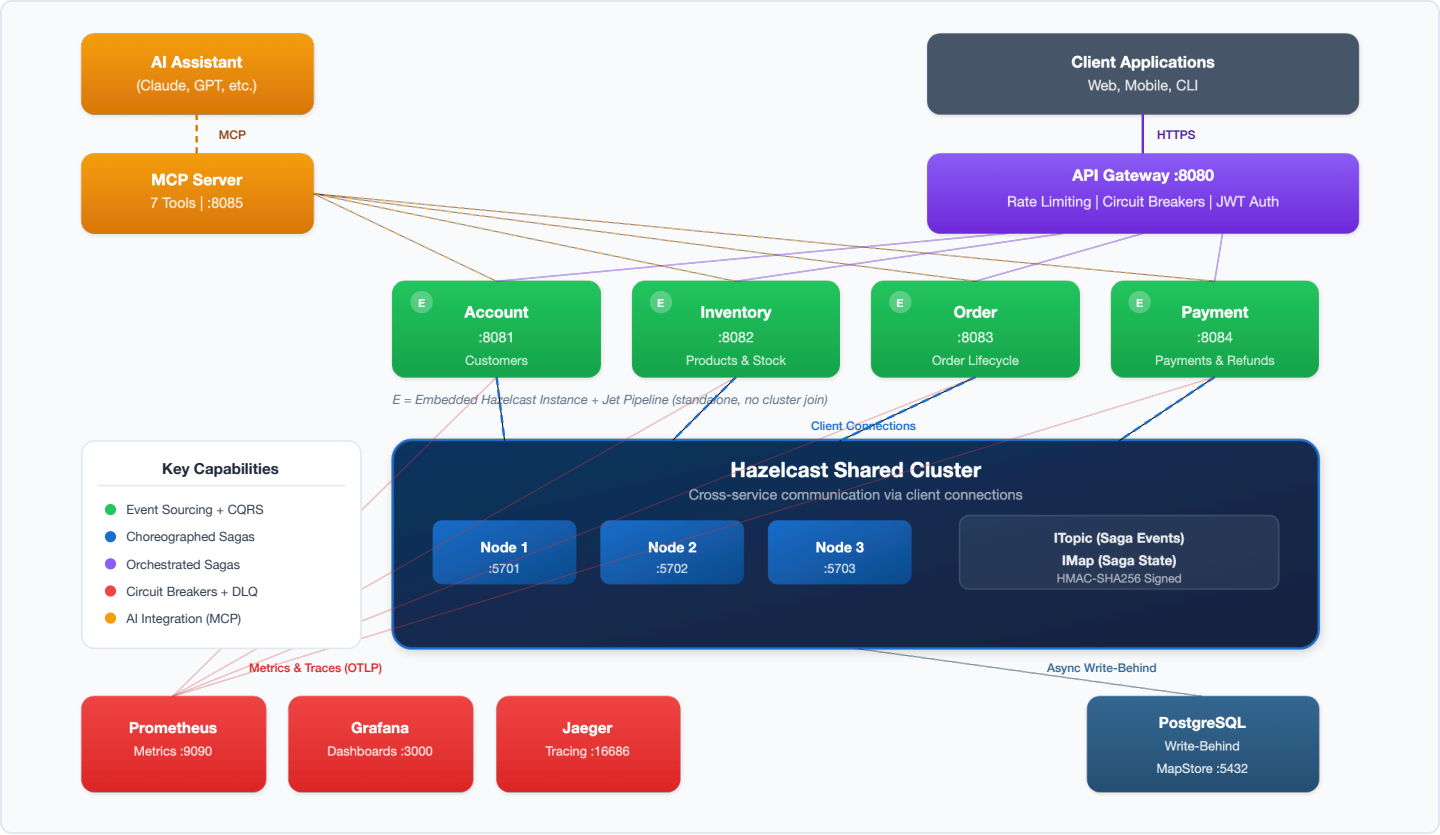
Microservices Framework

Production-ready event sourcing, saga coordination, and real-time stream processing — built entirely on Hazelcast and Spring Boot.

100K+	<1ms	54K+	12.6ms	10	4
VIEW OPS/SEC	P99 VIEW LATENCY	SAGAS EXECUTED	P50 E2E LATENCY	MCP AI TOOLS	MICROSERVICES

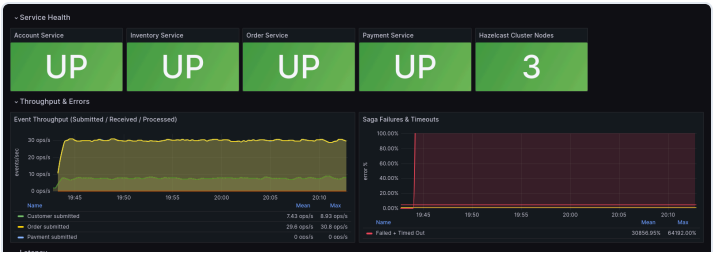
Hazelcast is a great platform for supporting microservice architectures. But for customers who aren't fully there yet, transitioning from a monolith to microservices is more than just breaking things apart — do it wrong and you end up with many tightly-coupled pieces where any failure breaks the whole system. Using microservice patterns effectively is the key. This demo and the accompanying series of blog posts walks prospects or customers through the right way to implement a Hazelcast-centric microservice architecture.

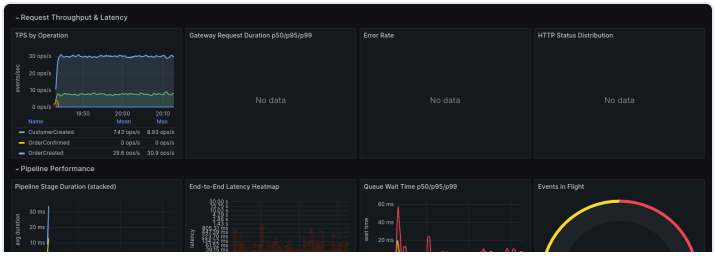
System Architecture



Dual-instance architecture: each service runs an embedded standalone Hazelcast (Jet pipelines) + client connection to shared 3-node cluster (cross-service sagas via ITopic)

Live Observability — Pre-Provisioned Dashboards





Capabilities & Technical Details

Event Sourcing & CQRS

Append-only event store backed by IMap with event journal. Materialized views rebuilt from events via Jet pipelines deliver 100K+ ops/sec at sub-millisecond latency.

Distributed Sagas

Both choreographed (event-driven, no coordinator) and orchestrated (central controller, HTTP steps) saga patterns with automatic compensation, timeout detection, and per-step metrics.

Resilience & Reliability

Resilience4j circuit breakers with retry on all saga listeners. Transactional outbox for guaranteed delivery. Dead letter queue with admin REST API. Idempotency guards for exactly-once processing.

AI-Powered Similarity Search

Hazelcast VectorCollection with HNSW indexing for product similarity. Supports cosine, dot-product, and euclidean metrics. `GET /api/products/{id}/similar`. Enterprise VectorCollection with HNSW indexing; Community fallback uses IMap with brute-force cosine similarity.

AI Integration (MCP)

10 MCP tools let AI assistants query views, submit events, inspect saga state, manage dead letter queues, check metrics, and run demos. Supports stdio and HTTP/SSE transport with API key RBAC.

Full Observability

Micrometer + Prometheus metrics across all services. Seven pre-provisioned Grafana dashboards including business metrics. Jaeger distributed tracing via OTLP. Per-step saga duration metrics.

Durable Persistence

Write-behind MapStore batches events and views to PostgreSQL asynchronously. Automatic cold-start reload. Bounded IMap memory with eviction. Provider-agnostic interfaces for any JDBC database.

Security (3 Layers)

JWT/OAuth2 on gateway & services. HMAC-SHA256 signing of ITopic saga events for service-to-service authentication. MCP API key RBAC (VIEWER / OPERATOR / ADMIN).

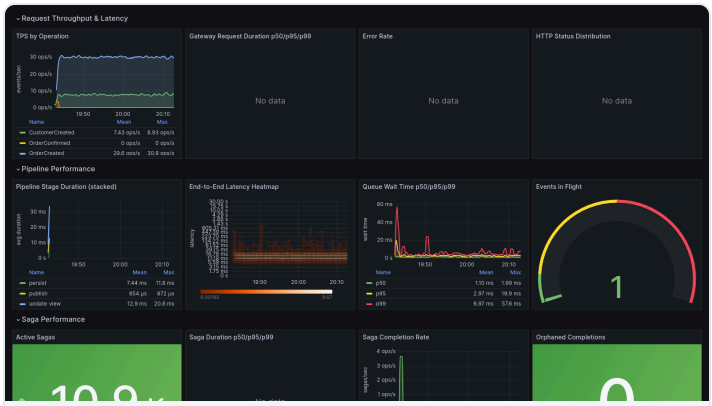
Deployment: Runs on a developer laptop via Docker Compose or deploys to AWS EKS with provided Helm charts and Terraform scripts. Other cloud providers (GCP GKE, Azure AKS) can be targeted by request.

Live Observability — Grafana Dashboard Suite

Seven pre-provisioned Grafana dashboards provide real-time visibility into every layer of the framework — from executive business metrics down to event flow, saga coordination, materialized view performance, and database persistence. Dashboards are auto-provisioned via Docker Compose and Helm charts — no manual setup required.

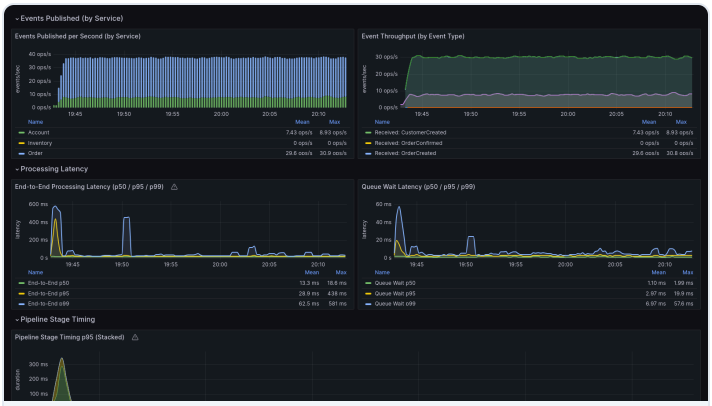
System Overview

Service health, JVM metrics, throughput, latency percentiles, error rates



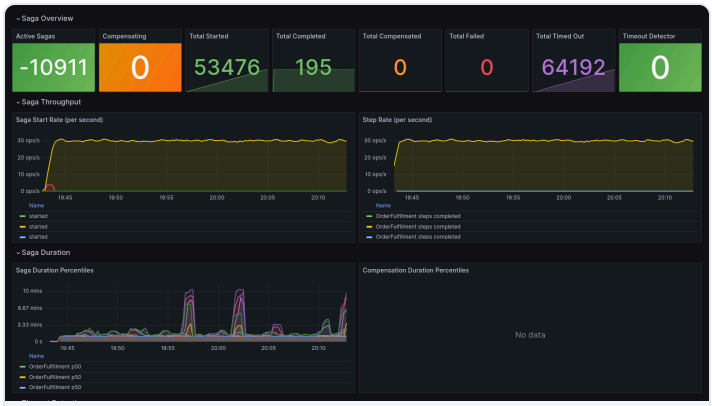
Performance Testing

TPS targets, pipeline throughput, saga completion rates, resource utilization



Event Flow

Event throughput, latency percentiles, pipeline stage timing, event types

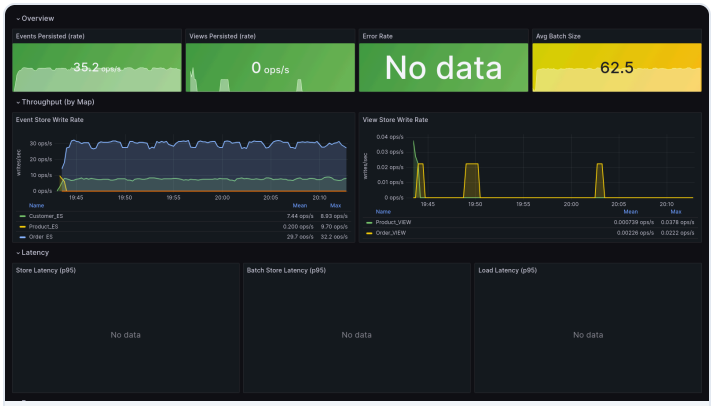


Saga Coordination

54K+ sagas, compensation tracking, step duration percentiles, timeout detection

Materialized Views

View rebuild latency, cache hit rates, entry counts, near cache performance



Persistence Layer

Write-behind queue depth, batch sizes, PostgreSQL latency, eviction metrics

Technology Stack

Component	Technology
Runtime	Java 17+
Framework	Spring Boot 3.2
Data Grid	Hazelcast 5.6 (CE default, EE opt-in)
Stream Processing	Hazelcast Jet
Database	PostgreSQL 16+ (write-behind)
Gateway	Spring Cloud Gateway
AI / ML	Spring AI MCP Server 1.0, VectorCollection
Metrics	Micrometer / Prometheus
Dashboards	Grafana 10.3
Tracing	Jaeger (OTLP)
Containers	Docker / Docker Compose / K8s Helm

Modules (10 Total)

framework-core	Domain-agnostic event sourcing engine
framework-enterprise	Enterprise Edition extensions (VectorCollection)
framework-postgres	Write-behind MapStore persistence
ecommerce-common	Shared domain objects, events, DTOs
account-service	Customer management (:8081)
inventory-service	Product catalog & stock (:8082)
order-service	Order lifecycle (:8083)
payment-service	Payments & refunds (:8084)
api-gateway	Routing, rate limiting, auth (:8080)
mcp-server	AI assistant integration (:8085)

Hazelcast Primitives Used

IMap (event store, views, saga state, outbox, DLQ) • **ITopic** (event bus) • **Jet Pipeline** (stream processing) • **Event Journal** (change stream) • **Entry Processor** (atomic updates) • **FlakeIdGenerator** (sequences) • **Near Cache** (query acceleration) • **MapStore** (write-behind persistence) • **VectorCollection** (similarity search, EE)

Accompanying Blog Series: A 13-part blog series walks through every architectural decision, from event sourcing fundamentals through saga patterns, persistence, AI integration, resilience, and production deployment. Each post maps to a working code milestone in the framework — ideal for customer-facing technical enablement or self-guided learning.

Get Started

Clone, build, and run the full four-service demo in under 5 minutes with Docker Compose. Pre-loaded sample data, demo scripts, and dashboards are ready out of the box. The framework is designed to be forked and extended for customer-specific demos and proof-of-concept engagements.

Repository: Available upon request
Quick Start: ./scripts/docker/start.sh
Demo: ./scripts/demo-scenarios.sh
Contact: mike.yawn@hazelcast.com